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Kempe-based HC(5): graphs with no K_6 minor are 5-colorable

Mader 1967: graphs w/ no K_6 minor have $e \leq 4n-10$.

Assume a maximally K_5 graph G. Given a vertex v in G, there exists 4 other vertices such that the 5 vertices form K_5 for all v in G.

~~• G is 5-colorable since G is 4-degenerate $\Rightarrow \chi(G) \leq k+1=5$.~~

1) $G = K_5$ has $\frac{n(n-1)}{2} = \frac{5(5-1)}{2} = 10$ edges

2) $G > K_5$ has $n-5$ added vertices from K_5, where each added ~~ed~~ vertex connects with 4 vertices.

$$e = 10 + (n-5) \cdot 4$$
$$= 10 + 4n - 20$$
$$= 4n - 10$$

$\therefore e = 4n-10$ for maximally K_5 graph.

• A maximally K_5 graph is 5-colorable, by induction. A maximally ~~planar~~ K_5 graph has no K_6 minor.

• ChatGPT says: degeneracy of maximally K_5 graph ≤ 5.

$$\begin{cases} \text{degeneracy (maximally } K_5 \text{ graph)} \leq 5 \\ \text{degeneracy (planar graph)} \leq 5 \\ \text{degeneracy (~~$K_{3,3}$ or~~ Wagner graph)} \leq 3 \end{cases}$$

A graph with no K_6 minor has a maximally planar K_5 network, a planar component (and Wagner component) joined by clique sums

~~degeneracy $\{\deg(G_1), \deg(G_2), \deg(G_3), k\}$~~

~~∴ a graph w/ no K_6 minor has degeneracy ≤ 5.~~

Graphs glued together along a k-clique has

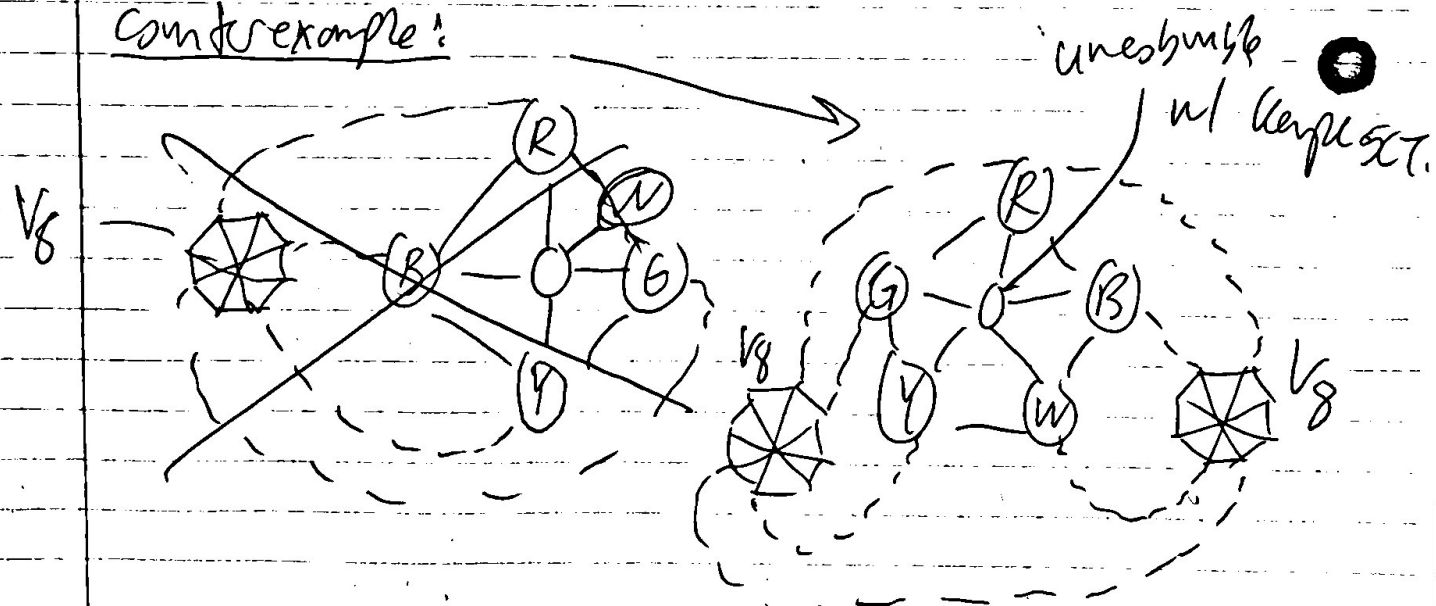
degeneracy $(G) \leq \max(\text{degeneracy}(G_1), \text{degeneracy}(G_2), k)$
– where k is k-clique sum.

For graphs w/ no K_6 minor consisting of maximally K_5, planar and wagner graphs

degeneracy [no K_6 minor graph] ≤ 5 $(k=5)$

∴ degeneracy $(G_{\text{no } K_6 \text{ minor}}) \leq 5$ try Kempe's method

<u>Counterexample:</u>



- Not colourable using Kempe's method due to $K_{3,3}$ ~~exp~~ i.e. Wagner V_8 and possibly K_5 ~~that~~ due to nonplanarity but still 5-degenerate.

- Next: will try a pre-coloured method that colours certain parts of the graph (K_5 & V_8) first, then planar part.